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**Ref. no: CRTC/CEE/G20220180/S- 33**

# SUB-SOIL INVESTIGATION REPORT

## FOR THE PROPOSED

### 5 STORIED RESIDENTIAL BUILDING

**AT BANIACHONG, HOBIGONJ.**
**DAG # 1101/1505, KHA # 1071, J.L # 113, MOUJA- TUPKHANA**

**CLIENT**

**ABDUL HANNAN**

Verification Code: 9Y1ns2Eu
Verify report at: crtc-cee-sust.com

**CONDUCTED BY**

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

**SAMPLE COLLECTED BY**

**SHAHIN CONSULTANT & ARCHITECT**
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## Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

## Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

## Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

**Table – 1: Items of the supplied samples**

| No | Name | Bore hole no | Depth | Quantities |
|---|---|---|---|---|
| 01 | Disturbed sample | 01 | 0-60ft | 12 |
| 02 | Disturbed sample | 02 | 0-50ft | 10 |
| 03 | Disturbed sample | 03 | 0-55ft | 11 |

## Laboratory Classifications

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

## Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

## Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

**Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)**

| N-values | Consistency | Unconfined Compressive Strength ($q_u$) | |
|---|---|---|---|
| | | tsf | kPa |
| 0-2 | Very soft | 0 to 0.25 | 0 to 25 |
| 3-5 | Soft | 0.25 to 0.50 | 25 to 50 |
| 6-9 | Medium | 0.50 to 1 | 50 to 100 |
| 10-16 | Stiff | 1 to 2 | 100 to 200 |
| 17-30 | Very stiff | 2 to 4 | 200 to 400 |
| >30 | Hard | >4 | >400 |

**Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 1 – 2 | Very Loose | 0 – 15 | 26 – 28 | 70 – 100 |
| 3 – 6 | Loose | 15 – 35 | 28 – 30 | 90 – 115 |
| 7 – 15 | Medium | 35 – 65 | 30 – 34 | 110 – 130 |
| 16 – 30 | Dense | 65 – 85 | 33 – 38 | 110 – 140 |
| >30 | Very Dense | >85 | >50 | 130 – 150 |

**Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 2 – 3 | Very Loose | 0 – 15 | 27 – 28 | 70 – 100 |
| 4 – 7 | Loose | 15 – 35 | 30 – 32 | 90 – 115 |
| 8 – 20 | Medium | 35 – 65 | 32 – 36 | 110 – 130 |
| 21 – 40 | Dense | 65 – 85 | 36 – 42 | 110 – 140 |

**Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 3 – 6 | Very Loose | 0 – 15 | 28 – 30 | 70 – 100 |
| 5 – 9 | Loose | 15 – 35 | 30 – 34 | 90 – 115 |
| 10 – 25 | Medium | 35 – 65 | 33 – 40 | 110 – 130 |
| 26 – 40 | Dense | 65 – 85 | 40 – 50 | 110 – 140 |
| >40 | Very Dense | >85 | >50 | 130 – 150 |

**Bearing capacity for deep foundation**

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

**Table 6: Formula Used to Determine Bearing Capacity of Soil**

| **Shallow Foundation** | |
|---|---|
| For Cohesive Soil | |
| $q_a = q_u + \gamma \frac{D}{3}$<br>$q_a = q_u$(Common Practice) | Where, $q_a$=Allowable bearing capacity, $q_u$= Unconfined Compressive Strength, $\gamma$= Unit Weight of soil, $D$= Depth of footing |
| For Cohesion less Soil | |
| $q_{ult} = 30N;$<br>$q_a = q_{ult}/FS$ | $q_{ult}$= Ultimate bearing capacity in kPa, $N$ = SPT Number, $FS$= Factor of Safety |
| **Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition** | |
| **Deep Foundation** | |
| For Cohesive Soil | |
| $f_{su} = \alpha C_u$ ;<br>$q_{pu} = N_c C_u$ | $\alpha$ = Adhesion factor, $f_{su}$ = Ultimate skin friction, $C_u$ = Undrained Shear Strength, $q_{pu}$ = Ultimate end bearing of pile, $N_c$ = Bearing capacity factor for deep foundation=9 |
| For Cohesion less Soil | |
| $f_{su} = 4N/200\ tsf$ ;<br>$q_{pu} = 4N\ tsf$ | $f_{su}$ = Ultimate skin friction, $q_{pu}$ = Ultimate end bearing of pile, $N$ = SPT Number |
| **Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"** | |

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

**Table 7**: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)**

| Bore Hole No. | Depth | Design N | Clay | | | | | | Sand |
|---|---|---|---|---|---|---|---|---|---|
| | | | Continuous | | | Square(Isolated) | | | |
| | (ft) | - | D/B=0.25 | D/B=0.5 | D/B=1 | D/B=0.25 | D/B=0.5 | D/B=1 | |
| BH01 | 2.5-5.0 | 16 | 1.902 | 2.001 | 2.181 | 2.247 | 2.427 | 2.640 | 2.050 |
| | 5.0-10 | 19 | 2.240 | 2.356 | 2.568 | 2.646 | 2.858 | 3.109 | 2.414 |
| BH02 | 2.5-5.0 | 17 | 1.949 | 2.050 | 2.234 | 2.302 | 2.486 | 2.705 | 2.100 |
| | 5.0-10 | 21 | 2.436 | 2.562 | 2.793 | 2.877 | 3.108 | 3.381 | 2.625 |
| BH03 | 2.5-5.0 | 17 | 2.018 | 2.123 | 2.314 | 2.384 | 2.575 | 2.801 | 2.175 |
| | 5.0-10 | 21 | 2.392 | 2.516 | 2.743 | 2.825 | 3.052 | 3.320 | 2.578 |
| BH04 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH05 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH06 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH07 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH08 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH09 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH10 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |

** Designer should consider both the calculated design N value and settlement behavior of the corresponding soil profile while selecting shallow footings

**Table 8: The allowable skin friction for precast driven piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.096 | 0.096 | 0.118 | - | - | - | - | - | - | - |
| 10 | 0.160 | 0.168 | 0.176 | - | - | - | - | - | - | - |
| 15 | 0.152 | 0.152 | 0.152 | - | - | - | - | - | - | - |
| 20 | 0.152 | 0.184 | 0.168 | - | - | - | - | - | - | - |
| 25 | 0.192 | 0.128 | 0.144 | - | - | - | - | - | - | - |
| 30 | 0.144 | 0.136 | 0.144 | - | - | - | - | - | - | - |
| 35 | 0.144 | 0.128 | 0.144 | - | - | - | - | - | - | - |
| 40 | 0.152 | 0.144 | 0.144 | - | - | - | - | - | - | - |
| 45 | 0.144 | 0.144 | 0.136 | - | - | - | - | - | - | - |
| 50 | 0.168 | 0.160 | 0.152 | - | - | - | - | - | - | - |
| 55 | 0.168 | - | 0.160 | - | - | - | - | - | - | - |
| 60 | 0.176 | - | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 9: The allowable end bearing for precast driven piles in Tsf (FS=3.0)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.900 | 0.900 | 1.125 | - | - | - | - | - | - | - |
| 10 | 32.000 | 33.600 | 35.200 | - | - | - | - | - | - | - |
| 15 | 30.400 | 30.400 | 30.400 | - | - | - | - | - | - | - |
| 20 | 30.400 | 36.800 | 33.600 | - | - | - | - | - | - | - |
| 25 | 38.400 | 25.600 | 28.800 | - | - | - | - | - | - | - |
| 30 | 28.800 | 27.200 | 28.800 | - | - | - | - | - | - | - |
| 35 | 28.800 | 25.600 | 28.800 | - | - | - | - | - | - | - |
| 40 | 30.400 | 28.800 | 28.800 | - | - | - | - | - | - | - |
| 45 | 28.800 | 28.800 | 27.200 | - | - | - | - | - | - | - |
| 50 | 33.600 | 32.000 | 30.400 | - | - | - | - | - | - | - |
| 55 | 33.600 | - | 32.000 | - | - | - | - | - | - | - |
| 60 | 35.200 | - | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.062 | 0.062 | 0.077 | - | - | - | - | - | - | - |
| 10 | 0.104 | 0.109 | 0.114 | - | - | - | - | - | - | - |
| 15 | 0.099 | 0.099 | 0.099 | - | - | - | - | - | - | - |
| 20 | 0.099 | 0.120 | 0.109 | - | - | - | - | - | - | - |
| 25 | 0.125 | 0.083 | 0.094 | - | - | - | - | - | - | - |
| 30 | 0.094 | 0.088 | 0.094 | - | - | - | - | - | - | - |
| 35 | 0.094 | 0.083 | 0.094 | - | - | - | - | - | - | - |
| 40 | 0.099 | 0.094 | 0.094 | - | - | - | - | - | - | - |
| 45 | 0.094 | 0.094 | 0.088 | - | - | - | - | - | - | - |
| 50 | 0.109 | 0.104 | 0.099 | - | - | - | - | - | - | - |
| 55 | 0.109 | - | 0.104 | - | - | - | - | - | - | - |
| 60 | 0.114 | - | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.405 | 0.405 | 0.506 | - | - | - | - | - | - | - |
| 10 | 14.400 | 15.120 | 15.840 | - | - | - | - | - | - | - |
| 15 | 13.680 | 13.680 | 13.680 | - | - | - | - | - | - | - |
| 20 | 13.680 | 16.560 | 15.120 | - | - | - | - | - | - | - |
| 25 | 17.280 | 11.520 | 12.960 | - | - | - | - | - | - | - |
| 30 | 12.960 | 12.240 | 12.960 | - | - | - | - | - | - | - |
| 35 | 12.960 | 11.520 | 12.960 | - | - | - | - | - | - | - |
| 40 | 13.680 | 12.960 | 12.960 | - | - | - | - | - | - | - |
| 45 | 12.960 | 12.960 | 12.240 | - | - | - | - | - | - | - |
| 50 | 15.120 | 14.400 | 13.680 | - | - | - | - | - | - | - |
| 55 | 15.120 | - | 14.400 | - | - | - | - | - | - | - |
| 60 | 15.840 | - | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

## CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed five storied residential building.

***Soil profile scenario***

Overall soil information of the investigated site is quite regular in between the boreholes. However, vertical inconsistency exists in the present soil formation. In all the three bore holes, clay soil of soft to medium consistency has been found in the top 5ft layer and then dense deposition of sand has been found up to the 50-60 ft depth. However, soil condition cannot be predicted beyond 60 ft depth since no sample is available from that depth.

***Foundation options***

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since the overall soil condition is quite regular in nature and dominated by sandy soil, shallow foundation could be a suitable option for the proposed structure. For designing, the necessary data can be found from the attached table **7, 8**, **9**, **10** and **11**.

***Safety provisions***

It is recommended to **check the settlement** of the site considering building load while designing either shallow or deep foundation. Obviously, **pile load test** (if provided) covering the entire project area must be carried out for confirming the design parameters of the service pile. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

***Attachments***

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. No other test has been done by the client.

***Special notes***

These suggestions are absolutely tentative, and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **50-60 ft** as client provided.

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